

On Regularization of Maximum Mean Discrepancy Gradient Flow

Problem: How to learn a target probability distribution π on $\mathbb{R}^d$.

- Sampling (e.g: $\pi \propto \exp(-V)$ is the posterior distribution in Bayesian inference).
- Generative models (e.g: π is the distribution of an image dataset).

Background: Optimization in the space of probability measures

Problem: How to learn a target probability distribution π on $\mathbb{R}^d$.

- This problem can be written as an **optimization** problem on $\mathcal{P}_2(\mathbb{R}^d)$.

$$\arg \min_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} D(\mu \mid \pi).$$

- Here D is a similarity metric or distance, e.g. Kullback–Leibler divergence.
 - $D(\mu \mid \pi) = 0$ if and only if $\mu = \pi$.
- $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of probability measures with a finite second moment.
 - We primarily consider $\mathcal{P}_2(\mathbb{R}^d)$ rather than $\mathcal{P}(\mathbb{R}^d)$ for the nice geometrical properties.
- How to find the minimum? Gradient descent!

Background: Euclidean gradient flow

- Euclidean gradient flow of an objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$

$$\partial_t x_t = -v(x_t), \quad v = \nabla F.$$

- ∇F denotes the gradient of F .
- This is the continuous-time analogue of gradient descent:

$$x_{n+1} = x_n - \gamma v(x_n),$$

where $\gamma > 0$ is the step size.

- When F is both **strongly convex and smooth**, Euclidean gradient **descent** converges exponentially fast [BV04].

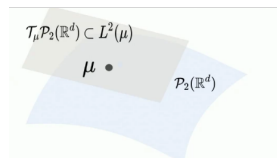
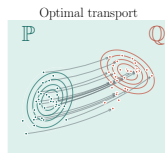
Background: Wasserstein gradient flow

Challenge: How to find $\arg \min_{\mu \in \mathcal{P}_2(\mathbb{R}^d)} D(\mu \mid \pi)$?

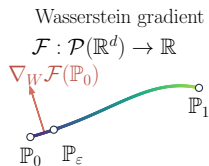
- Gradient descent! Wait... How do we define gradients in $\mathcal{P}_2(\mathbb{R}^d)$?
- Endow $\mathcal{P}_2(\mathbb{R}^d)$ with the Wasserstein-2 distance W_2 .

$$W_2^2(\nu, \mu) = \int \|T(x) - x\|^2 d\nu(x) = \|T - \text{Id}\|_{L_2(\nu)}^2.$$

- $W_2^2(\nu, \mu)$ means the **minimal energy** takes to transport mass from ν to μ .
- $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the **optimal transport map** from ν to μ .
- $(\mathcal{P}_2, W_2)$ can be 'treated' as a Riemann manifold under Otto's calculus [Ott01].
- The **tangent space** $\mathcal{T}_\mu \mathcal{P}_2(\mathbb{R}^d)$ at $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is a dense subset of $L_2(\mu)$.



Background: Wasserstein gradient flow



Definition (Wasserstein gradient)

Let $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$. The **Wasserstein gradient of $\mathcal{F}$** evaluated at $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ is the unique function $\nabla_{W_2} \mathcal{F}(\mu) : \mathbb{R}^d \rightarrow \mathbb{R}^d$, s.t. for any $T \in \mathcal{T}_\mu \mathcal{P}_2(\mathbb{R}^d)$,

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [\mathcal{F}((\text{Id} + \epsilon T)_\# \mu) - \mathcal{F}(\mu)] = \langle \nabla_{W_2} \mathcal{F}, T \rangle_{L_2(\mu)}.$$

- The gradient is defined along a ‘curve’ $(\text{Id} + \epsilon T)_\# \mu$ in $\mathcal{P}_2(\mathbb{R}^d)$.

Background: Wasserstein gradient flow

Definition (Wasserstein gradient flow)

Wasserstein gradient flow of $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$ is a sequence of distributions $(\mu_t)_{t \geq 0}$ that evolves according to the **continuity equation** $\partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0$, where the vector field $v_t = -\nabla_{W_2} \mathcal{F}(\mu_t)$. Or equivalently, $\dot{x}_t = v_t(x_t)$ with $\mu_t = \text{Law}(x_t)$.

Euclidean Gradient Flow

- State space: $\mathbb{R}^d$
- Objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$.
- Update scheme: $\dot{x}_t = v_t(x_t)$, with $v_t = -\nabla F$.

Wasserstein Gradient Flow

- State space: $\mathcal{P}_2(\mathbb{R}^d)$
- Objective $\mathcal{F} : \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}$:
- Update scheme $\dot{x}_t = v_t(x_t)$, with $v_t = -\nabla_{W_2} \mathcal{F}(\mu_t)$.

Example 1: Langevin diffusion [JKO98]

- Given the target distribution $\pi \propto \exp(-V)$ with $V : \mathbb{R}^d \rightarrow \mathbb{R}$.
- The functional $\mathcal{F}_{\text{KL}} = \text{KL}(\cdot, \pi)$ and its Wasserstein gradient

$$[\nabla_{W_2} \mathcal{F}_{\text{KL}}(\mu)](\cdot) = \nabla V(\cdot) + \nabla \log \mu_t(\cdot).$$

- The Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

$$\partial_t \mu_t = \nabla \cdot (\mu_t (\nabla V + \nabla \log \mu_t)).$$

- It is equivalent to the Fokker–Planck equation of the **Langevin diffusion** [SS19]:

$$\mathrm{d}x_t = -\nabla V(x_t) \mathrm{d}t + \sqrt{2} \mathrm{d}W_t, \quad \mu_t = \text{Law}(x_t).$$

- A standard time-discretization (Euler–Maruyama scheme) is:

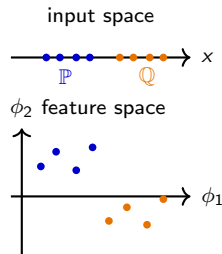
$$x_{n+1} = x_n - \gamma \nabla V(x_n) + \sqrt{2\gamma} \xi_n, \quad \xi_n \sim \mathcal{N}(0, I_d).$$

Background: Maximum Mean Discrepancy (MMD) [GBR⁺12]

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



Theorem 1

MMD admits a closed-form expression

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian) [SFL11].

Example 2: MMD gradient flow [AKSG19]

- The functional $\mathcal{F}_{\text{MMD}} = \frac{1}{2}\text{MMD}^2(\cdot, \pi)$
- Its Wasserstein gradient

$$[\nabla_{W_2}\mathcal{F}_{\text{MMD}}(\mu)](\cdot) = \nabla\left(\int k(x, \cdot) \, \mathrm{d}\mu(x) - \int k(x, \cdot) \, \mathrm{d}\pi(x)\right).$$

- The Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$,

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_t)), \quad \mathrm{d}x_t = -[\nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_t)](x_t) \, \mathrm{d}t.$$

- A standard time-discretization (Euler–Maruyama scheme) is:

$$x_{n+1} = x_n - \gamma \nabla_{W_2} \mathcal{F}_{\text{MMD}}(\mu_n)(x_n), \quad \text{Law}(x_n) = \mu_n.$$

- The same holds for $\mathcal{F}_{\text{KSD}} = \frac{1}{2}\text{MMD}^2(\cdot, \pi)$ with a Stein kernel k_π [KAFMA21].

Wasserstein gradient flow of $\mathcal{F}_{\text{KL}}$

- $\pi \propto \exp(-V)$
- Sampling
- Convergence in discrete time

$$\text{KL}(\mu_n, \pi) \leq \exp(-\alpha\gamma n) \text{KL}(\mu_n, \pi) + \frac{\gamma n \beta^2}{\alpha}.$$

- β is the Lipschitz continuity of V .
- $\lim_{n \rightarrow \infty} \text{KL}(\mu_n, \pi) = 0!$.

Wasserstein gradient flow of $\mathcal{F}_{\text{MMD}}$

- $\{y_i\}_{i=1}^M$ i.i.d samples from π
- Generative modelling
- **Non**-convergence in discrete time

$$\text{MMD}^2(\mu_n, \pi) \leq \frac{W_2^2(\mu_0, \pi)}{\gamma n} + \bar{K}.$$

- $\bar{K}$ is a positive barrier term that does not vanish.
- $\lim_{n \rightarrow \infty} \text{MMD}(\mu_n, \pi) \neq 0!$

Non-convexity of MMD prevents global convergence

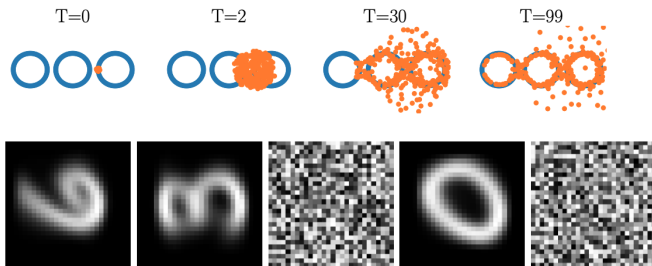


Figure: [BSM25]

- Convergence of MMD flow has been proved under noise injection and stringent conditions on the noise scale [AKSG19, CKK⁺25].
- Disclaimer: There exists many papers where MMD flow **empirically** generate high-quality images with adversarial training of kernels [GBG25], or with non-smooth kernels plus deep network distillation [HWAH24, AHS23].

Non-convexity of MMD prevents global convergence

Question: How can we overcome the non-convexity of $\mathcal{F}_{\text{MMD}}$?

- In this talk, I will present three complementary routes.

1. De-regularized MMD gradient flow

arXiv:2409.14980

2. Sobolev-regularized MMD gradient flow

arXiv:2605.11884

3. Applications: 1) Statistical inference; 2) Generative model

arXiv:2607.03871, *Work in progress*

DrMMD flow and SrMMD flow



Zonghao
Chen



Aratrika
Mustafi



Pierre
Glaser



Anna
Korba



Arthur
Gretton



Bharath K.
Sriperumbudur

Kernel assumptions

Assumption 1 (Growth I)

There exists a constant $M > 0$ such that, for every $x \in \mathbb{R}^d$,

$$\begin{aligned} |k(x, x)| &\leq M (1 + |x|^4), \\ |\partial_{1,i} \partial_{2,i} k(x, x)| &\leq M (1 + |x|^2), \quad i \in [d]. \end{aligned}$$

Assumption 2 (Local Boundedness)

For $R > 0$, let $\mathcal{K} = B(0, R) \subset \mathbb{R}^d$. There exists a $B_R > 0$ such that, for every $i, j \in [d]$,

$$\sup_{x \in \mathcal{K}} |\partial_{1,i} \partial_{1,j} \partial_{2,i} \partial_{2,j} k(x, x)| \leq B_R.$$

- Gaussian, Matérn, and inverse multiquadratic kernels.
- Polynomial kernels.
- Stein kernels with score functions of at most linear growth.

DrMMD: motivation

$T_\pi : L_2(\pi) \rightarrow L_2(\pi)$ is the kernel **integral operator** defined as

$$T_\pi f(\cdot) = \int k(x, \cdot) f(x) \, d\pi(x).$$

Proposition 1 (MMD and χ^2 -divergence)

Suppose μ is absolutely continuous with respect to π , i.e., $\mu \ll \pi$. Then

$$\text{MMD}^2(\mu, \pi) = \left\| T_\pi^{\frac{1}{2}} \left(\frac{d\mu}{d\pi} - 1 \right) \right\|_{L_2(\pi)}^2 \quad \text{and} \quad \chi^2(\mu, \pi) = \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L_2(\pi)}^2.$$

- Similar to $\mathcal{F}_{\text{KL}}$, $\mathcal{F}_{\chi^2}(\cdot) = \chi^2(\cdot, \pi)$ flow enjoys convergence for Gaussian-like π [OT11].
- An **interpolation** between MMD^2 and χ^2 ?

DrMMD: An interpolation of MMD and χ^2

Definition (De-regularized Maximum Mean Discrepancy (DrMMD))

Suppose $\mu \ll \pi$ where $\mu, \pi \in \mathcal{P}_2(\mathbb{R}^d)$. Then the (de)-regularized maximum mean discrepancy (DrMMD) between μ, π is defined as, for $\lambda > 0$,

$$\text{DrMMD}(\mu, \pi) = (1 + \lambda) \left\| \left((\mathcal{T}_\pi + \lambda \text{Id})^{-1} \mathcal{T}_\pi \right)^{\frac{1}{2}} \left(\frac{d\mu}{d\pi} - 1 \right) \right\|_{L_2(\pi)}^2.$$

Proposition 2 (Interpolation of MMD^2 and χ^2)

$$\lim_{\lambda \rightarrow 0} \text{DrMMD}(\mu, \pi) = \chi^2(\mu, \pi), \quad \lim_{\lambda \rightarrow \infty} \text{DrMMD}(\mu, \pi) = \text{MMD}^2(\mu, \pi).$$

- Similar ideas in kernel hypothesis testing [MRW⁺99, HBM07, HKSL24].
- Interpolation between MMD and general f-divergences [GAG21, SNRS26].

DrMMD gradient flow

Definition ($\mathcal{F}_{\text{DrMMD}}$ gradient flow)

The Wasserstein gradient flow of $\mathcal{F}_{\text{DrMMD}}(\cdot) = \text{DrMMD}(\cdot, \pi)$ is $(\mu_t)_{t \geq 0}$ with

$$\partial_t \mu_t + \nabla \cdot (\mu_t (1 + \lambda) \nabla h_{\mu_t, \pi}) = 0.$$

where $h_{\mu, \pi} = (T_\pi + \lambda I)^{-1} T_\pi \left(\frac{d\mu}{d\pi} - 1 \right)$.

- Implementation: $h_{\mu, \pi} = (T_\pi + \lambda I)^{-1} (m_\mu - m_\pi)$ with m_μ, m_π the mean embeddings.
- MMD goes to grad school, becomes DrMMD, then drives a new Wasserstein gradient flow.



DrMMD: Convergence of $\mathcal{F}_{\chi^2}$ flow

Definition (Poincaré inequality)

We say that π satisfies a **Poincaré inequality** with constant C_P if for all $f, \nabla f \in L_2(\pi)$,

$$\text{Var}_\pi[f] \leq C_P \mathbb{E}_\pi [\|\nabla f\|^2].$$

- By “Gaussian-like,” we mean that π satisfies a Poincaré inequality, which in particular implies **sub-exponential tails** [BGL⁺14].
- For example, if $\pi = \mathcal{N}(0, a \cdot \text{Id})$, then $C_P = a^{-1}$.

Proposition 3 (Convergence of $\mathcal{F}_{\chi^2}$ flow)

Suppose that π satisfies the Poincaré inequality. Let $(\mu_t)_{t \geq 0}$ follow the $\mathcal{F}_{\chi^2}$ flow. Then, for any $T \geq 0$, $\text{KL}(\mu_T, \pi) \leq \exp(-2C_P^{-1}T) \text{KL}(\mu_0, \pi)$.

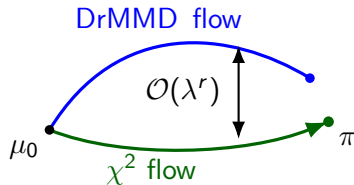
DrMMD: Convergence of DrMMD gradient flow

Theorem 2 (Convergence of DrMMD gradient flow)

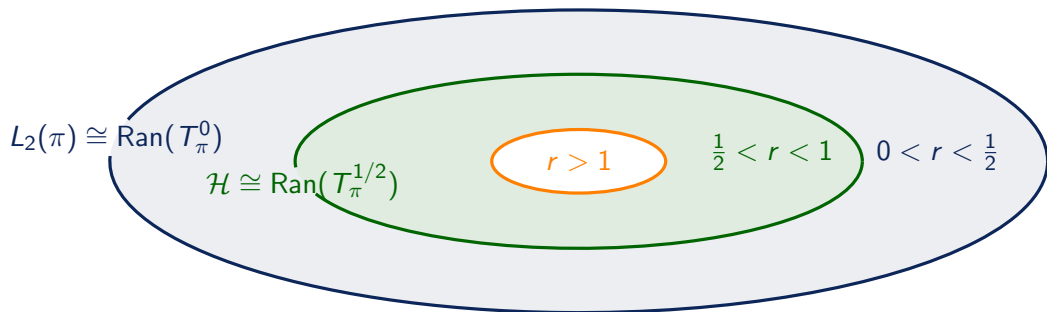
Let $(\mu_t)_{t \geq 0}$ be the $\mathcal{F}_{\text{DrMMD}}$ flow. Suppose the target distribution π satisfies a Poincaré inequality with constant C_P . Suppose that $\frac{d\mu_t}{d\pi} - 1 \in \text{Ran}(T_\pi^r)$ with $r > 0$. Then for any $T \geq 0$,

$$\text{KL}(\mu_T, \pi) \leq \exp\left(-\frac{2(1+\lambda)}{C_P} T\right) \text{KL}(\mu_0, \pi) + \mathcal{O}(\lambda^r C_P).$$

- Larger r means more regularity of the trajectory and thus smaller bias.
- Smaller Poincaré means faster convergence.
- For continuous-time DrMMD flow, we want $\lambda \rightarrow 0$ for 'convexity'; this is not the case for discrete time.



DrMMD: Interpolation space $\text{Ran}(T_\pi^r)$



- Large r indicates larger smoothness.

- DrMMD gradient flow (continuity equation)

$$\partial_t \mu_t + \nabla \cdot (\mu_t (1 + \lambda) \nabla h_{\mu_t, \pi}) = 0$$

- DrMMD gradient descent: for step size $\gamma > 0$,

$$\mu_{n+1} = (\text{Id} + \gamma(1 + \lambda) \nabla h_{\mu_n, \pi})_{\#} \mu_n.$$

DrMMD: Convergence of DrMMD gradient descent

Proposition 4 (Descent lemma of DrMMD gradient descent)

Let $(\mu_n)_{n \in \mathbb{N}}$ be the Wasserstein gradient descent of $\mathcal{F}_{\text{DrMMD}}$. Suppose $\pi \propto \exp(-V)$ with $\mathbf{H}V \preceq \beta \mathbf{I}$. Suppose the step size γ is small enough.

$$\text{KL}(\mu_{n+1}, \pi) - \text{KL}(\mu_n, \pi) \leq -\frac{2}{C_p} \chi^2(\mu_n, \pi) \gamma + \underbrace{\gamma \lambda^r}_{\text{Approximation error}} + \underbrace{\gamma^2 \beta \chi^2(\mu_n, \pi) \frac{1}{\lambda}}_{\text{Discretization error}}$$

- A trade-off on λ between approximation error and time-discretization error.
- Optimal choice of adaptive λ_n at each iterate n :

$$\lambda_n \propto \chi^2(\mu_n, \pi)^{\frac{1}{r+1}}$$

- At the start, a larger λ for more smoothness; closer to the convergence, a smaller λ to operate in the χ^2 regime to better catch the difference.

DrMMD: Convergence of DrMMD gradient descent

Theorem 3 (Convergence of DrMMD gradient descent)

Let $(\mu_n)_{n \in \mathbb{N}}$ be the Wasserstein gradient descent of $\mathcal{F}_{\text{DrMMD}}$. Suppose all conditions from the descent lemma hold. Then, for any $n_{\max} \in \mathbb{N}$,

$$\text{KL}(\mu_{n_{\max}}, \pi) \leq \exp\left(-\frac{2n_{\max}\gamma}{C_P}\right) \text{KL}(\mu_0, \pi) + \mathcal{O}(\gamma^{\frac{r}{r+1}} C_P \beta^{\frac{r}{r+1}}).$$

- To reach error $\text{KL}(\mu_{n_{\max}}, \pi) \leq \delta$, it takes $\mathcal{O}((\frac{1}{\delta})^{\frac{r+1}{r}} \log \frac{1}{\delta})$ iterations.
- By comparison, for Langevin Monte Carlo, it takes $\mathcal{O}(\frac{1}{\delta} \log \frac{1}{\delta})$ [CEL+24].
- DrMMD gradient descent takes more iterations due to the additional approximation error $\mathcal{O}(\lambda^r)$, but it **operates without the knowledge of potential V** and only requires $\Sigma_\pi = \int k(x, \cdot) \otimes k(x, \cdot) d\pi$ and the embedding $\int k(x, \cdot) d\pi$.

SrMMD: motivation

- Recall DrMMD.

$$\text{DrMMD}(\mu, \pi) = \sup_{f \in \mathcal{H}: \|f\|_{L_2(\mu)}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mu - \pi).$$

Definition (Sobolev-regularized Maximum Mean Discrepancy (SrMMD))

$$\text{SrMMD}(\mu, \pi) = \sup_{f \in \mathcal{H}: \|\nabla f\|_{L_2^d(\mu)}^2 + \lambda \|f\|_{\mathcal{H}}^2 \leq 1} \int f \, d(\mu - \pi).$$

- This was first proposed in [MSR19].

SrMMD: motivation

- Define $S_\mu : L_2(\mu) \rightarrow L_2(\mu)$ as the **gradient covariance** operator:

$$S_\mu f(\cdot) = \int \nabla f(x)^\top \nabla_1 k(x, \cdot) d\mu(x).$$

- Define $m_\mu = \int k(x, \cdot) d\mu(x)$ as the kernel mean embedding.

Definition (SrMMD gradient flow)

The SrMMD flow is $(\mu_t)_{t \geq 0}$ that satisfies

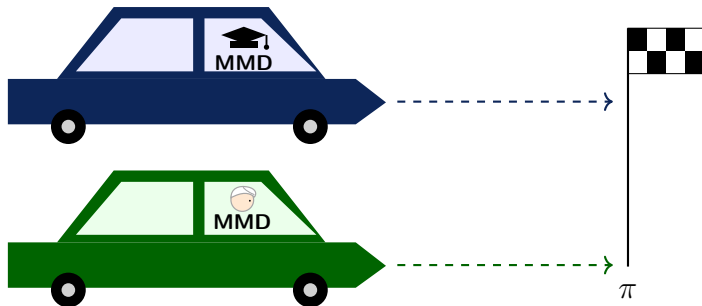
$$\partial_t \mu_t + \nabla \cdot (\mu_t (1 + \lambda) \nabla f_{\mu_t, \pi}) = 0.$$

where $f_{\mu, \pi} = (S_\mu + \lambda I)^{-1} (m_\mu - m_\pi)$.

- For comparison, $f_{\mu, \pi} = (T_\pi + \lambda I)^{-1} (m_\mu - m_\pi)$ for DrMMD flow.

SrMMD: SrMMD gradient flow

- MMD gets old: becomes SrMMD, then drives a new Wasserstein gradient flow.



SrMMD: Convergence

Theorem 4 (Convergence of SrMMD flow)

Let $(\mu_t)_{t \geq 0}$ be the SrMMD flow. Suppose that there exists $0 < \alpha \leq 1$ such that $m_{\mu_t} - m_\pi \in \text{Ran}(S_{\mu_t}^{\alpha/2})$ for all t . Then, for any $T \geq 0$,

$$\text{MMD}^2(\mu_T, \pi) \leq \exp(-2T) \text{MMD}^2(\mu_0, \pi) + \mathcal{O}(\lambda^\alpha).$$

Theorem 5 (Convergence of SrMMD descent)

Let $(\mu_n)_{n \in \mathbb{N}}$ be the SrMMD descent. Suppose that there exists $0 < \alpha \leq 1$ such that $m_{\mu_n} - m_\pi \in \text{Ran}(S_{\mu_n}^{\alpha/2})$ for all n . Take $\lambda \propto d\gamma$. Then, for any $n_{\max} \geq 0$,

$$\text{MMD}^2(\mu_{n_{\max}}, \pi) \leq \exp(-2n_{\max}\gamma) \text{MMD}^2(\mu_0, \pi) + \mathcal{O}(\gamma^\alpha).$$

- $\text{Ran}(S_\mu^{\alpha/2}) \cong H^{r(\alpha+1)-\alpha}$ on a torus.
- $\lambda \propto d\gamma$ balances smoothness error $\mathcal{O}(\lambda^\alpha \gamma)$ and discretization error $\mathcal{O}(\lambda^{\alpha-1} \gamma^2)$.

SrMMD: Iteration complexity

- Number of iterations sufficient to reach δ -closeness to π .

Algorithm	Metric	Iteration complexity
LMC [CEL ⁺ 24, VW19]	KL / χ^2	$\tilde{\mathcal{O}}(\delta^{-1}\beta)$
SVGD [BBG25]	KSD ²	$\mathcal{O}(\delta^{-1})$
R-SVGD [HBSL24]	Fisher	$\mathcal{O}(\delta^{-1})$
DrMMD	KL	$\tilde{\mathcal{O}}(\delta^{-(\alpha+1)/\alpha}\beta)$
SrMMD	MMD ² / KSD ²	$\tilde{\mathcal{O}}(\delta^{-1/\alpha})$

Table: Here, β denotes the isoperimetric constant of π .
 α denotes the regularity parameter of the flow.

- Different metrics: χ^2 stronger than KL and both are stronger than MMD/KSD [GM17, KBGM25].
- SrMMD rate is **independent** of isoperimetric constant of π .

In the following, we turn to applications

1) Minimum MMD estimation 2) Generative models



François-Xavier
Briol



Sophia
Kang



Louis
Sharrock



Arthur
Gretton

Application: Statistical inference

- $\mathbb{P}_\theta$ is the model; and $\mathbb{Q}$ is the data distribution.
- Maximum Likelihood estimator: $\arg \min_{\theta \in \Theta} \text{KL}(\mathbb{Q}, \mathbb{P}_\theta)$.
- Minimum MMD estimator [BBDG19, CAA22, ACADF23a, AG25]

$$\begin{aligned}\theta_{\text{MMD}} &= \arg \min_{\theta \in \Theta} \text{MMD}^2(\mathbb{P}_\theta, \mathbb{Q}) \\ &= \arg \min_{\theta \in \Theta} \left\{ \mathbb{E}_{x, x' \sim \mathbb{P}_\theta} [k(x, x')] - 2\mathbb{E}_{x \sim \mathbb{P}_\theta, y \sim \mathbb{Q}} [k(x, y)] + \text{const} \right\}.\end{aligned}$$

- Applications: [KGBF25, SKA⁺23, BFM25, FAZL23, AG24, AFP26, DOD25, DKDB22, DD23, CDB25, SK25, ACADF23b, CAN25, KPLB26].

Advantage 1: Likelihood-free inference?

✓ Only requires samples from $\mathbb{P}_\theta$ and $\mathbb{Q}$.

Advantage 2: Robustness to outliers?

✓ Bounded effects under bounded kernels.

Application: Statistical inference

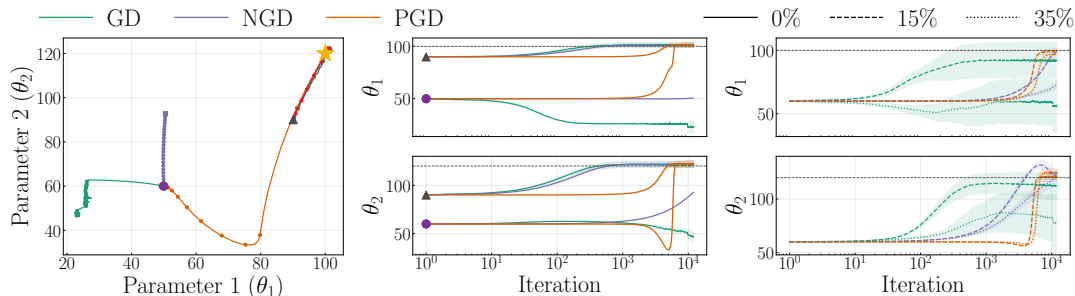
Question: How can we reliably find θ_{MMD} ?

1. MMD gradient flow $\mathbb{P}_{t+1} = (\text{Id} - \gamma \nabla f_{\mathbb{P}_t, \mathbb{Q}})_{\#} \mathbb{P}_t$.
2. Consider the model family: $\mathbb{P}_{\theta} = (T_{\theta})_{\#} \rho$: $\mathbb{P}_{\theta_{t+1}} = (T_{\theta_t - \gamma \Delta \theta_t})_{\#} \rho$.
3. Choose $\mathbb{P}_{\theta_{t+1}}$ closest to the MMD flow iterate $\mathbb{P}_{t+1}$.

$$\Delta \theta_t^{\text{PGD}} = \underbrace{\left\{ \mathbb{E}_{z \sim \rho} [\mathbf{J}_{\theta} G_{\theta_t}(z)^{\top} \mathbf{J}_{\theta} G_{\theta_t}(z)] \right\}^{-1}}_{\text{preconditioner}} \underbrace{\nabla_{\theta} \text{MMD}^2(\mathbb{P}_{\theta_t}, \mathbb{Q})}_{\text{standard gradient}}.$$

- Convergence: $\lim_{t \rightarrow \infty} \text{MMD}(\mathbb{P}_{\theta_t}, \mathbb{Q}) = \min_{\theta} \text{MMD}(\mathbb{P}_{\theta}, \mathbb{Q})$.

Experiment: Lotka–Volterra inference



- We compare gradient descent (GD), natural gradient descent (NGD) [BBDG19], and preconditioned gradient descent (PGD, ours).
- First row: Close initialization. Second row: Far initialization.

Experiment: composite goodness-of-fit testing [KGBF25]

Method	Type-I error ($\downarrow$)	Time	Power ($\uparrow$)	Time
PGD ($\mathcal{I} = 1$)	0.050 ± 0.018	8.6s	0.829 ± 0.032	3.6s
GD ($\mathcal{I} = 400$)	0.107 ± 0.026	10.4s	0.600 ± 0.041	4.9s
GD ($\mathcal{I} = 1$)	0.071 ± 0.022	2.6s	0.279 ± 0.038	1.1s

Table: Composite goodness-of-fit test. $\mathcal{I}$ represents the number of restarts.

Application: Generative models



wing



hammerhead shark



Irish terrier



coyote



schooner



water buffalo



manhole cover



fire engine



vending machine



dining table

- ResNet-style encoder pre-trained with masked autoencoding (MAE) [DLL⁺26].

- Minimization of MMD

About Me



- Zonghao Chen
- 4th year PhD Student at University College London (UCL)
 - Foundational AI Centre
 - Gatsby Computational Neuroscience Unit
- Graduated from Tsinghua University in 2022
 - Department of EE
- Kernel (nonparametric) methods, causal inference, statistical learning theory

- [ACADF23a] Pierre Alquier, Badr-Eddine Chérif-Abdellatif, Alexis Derumigny, and Jean-David Fermanian. Estimation of copulas via maximum mean discrepancy. *Journal of the American Statistical Association*, 118(543):1997–2012, 2023.
- [ACADF23b] Pierre Alquier, Badr-Eddine Chérif-Abdellatif, Alexis Derumigny, and Jean-David Fermanian. Estimation of copulas via maximum mean discrepancy. *Journal of the American Statistical Association*, 118(543):1997–2012, 2023.
- [AFP26] Pierre Alquier, Jean-David Fermanian, and Benjamin Poignard. Estimation of time series by maximum mean discrepancy. *arXiv preprint arXiv:2601.11233*, 2026.
- [AG24] Pierre Alquier and Mathieu Gerber. Universal robust regression via maximum mean discrepancy. *arXiv preprint arXiv:2006.00840*, 2024.
- [AG25] Pierre Alquier and Mathieu Gerber. regMMD: A package for parametric estimation and regression with maximum mean discrepancy. *arXiv preprint arXiv:2503.05297*, 2025.
- [AHS23] Fabian Altekrüger, Johannes Hertrich, and Gabriele Steidl. Neural

wasserstein gradient flows for maximum mean discrepancies with riesz kernels. *arXiv preprint arXiv:2301.11624*, 2023.

- [AKSG19] Michael Arbel, Anna Korba, Adil Salim, and Arthur Gretton. Maximum mean discrepancy gradient flow. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.
- [BBDG19] Francois-Xavier Briol, Alessandro Barp, Andrew B Duncan, and Mark Girolami. Statistical inference for generative models with maximum mean discrepancy. *arXiv preprint arXiv:1906.05944*, 2019.
- [BBG25] Sayan Banerjee, Krishna Balasubramanian, and Promit Ghosal. Improved finite-particle convergence rates for Stein variational gradient descent. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [BFM25] Florian Brück, Jean-David Fermanian, and Aleksey Min. Distribution free MMD tests for model selection with estimated parameters. *arXiv preprint arXiv:2305.07549*, 2025.
- [BGL⁺14] Dominique Bakry, Ivan Gentil, Michel Ledoux, et al. *Analysis and geometry of Markov diffusion operators*, volume 103. Springer, 2014.

- [BSM25] Ayoub Belhadji, Daniel Sharp, and Youssef Marzouk. Weighted quantization using mmd: From mean field to mean shift via gradient flows. *arXiv preprint arXiv:2502.10600*, 2025.
- [BV04] Stephen P. Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.
- [CAA22] Badr-Eddine Ch'erief-Abdellatif and Pierre Alquier. Finite sample properties of parametric mmd estimation: robustness to misspecification and dependence. *Bernoulli*, 28(1):181–213, 2022.
- [CAN25] Badr-Eddine Ch'erief-Abdellatif and Jeffrey Näf. Parametric MMD estimation with missing values: Robustness to missingness and data model misspecification. *arXiv preprint arXiv:2503.00448*, 2025.
- [CDB25] Zonghao Chen, Charita Dellaporta, and François-Xavier Briol. Total variation generalized Bayesian inference. *arXiv preprint*, 2025.
- [CEL⁺24] Sinho Chewi, Murat A. Erdogdu, Mufan Li, Ruqi Shen, and Matthew S. Zhang. Analysis of Langevin Monte Carlo from Poincare to log-Sobolev. *Foundations of Computational Mathematics*, pages 1–51, 2024.
- [CKK⁺25] Zonghao Chen, Toni Karvonen, Heishiro Kanagawa, François-Xavier Briol, and Chris Oates. Stationary MMD points for cubature. 2025.

- [DD23] Charita Dellaporta and Theodoros Damoulas. Robust Bayesian inference for Berkson and classical measurement error models. *arXiv preprint arXiv:2306.01468*, 2023.
- [DKDB22] Charita Dellaporta, Jeremias Knoblauch, Theodoros Damoulas, and François-Xavier Briol. Robust Bayesian inference for simulator-based models via the MMD posterior bootstrap. *arXiv preprint arXiv:2202.04744*, 2022.
- [DLL⁺26] Mingyang Deng, He Li, Tianhong Li, Yilun Du, and Kaiming He. Generative modeling via drifting. *arXiv preprint arXiv:2602.04770*, 2026.
- [DOD25] Charita Dellaporta, Patrick O’Hara, and Theodoros Damoulas. Decision making under model misspecification: DRO with robust Bayesian ambiguity sets. *arXiv preprint arXiv:2505.03585*, 2025.
- [FAZL23] Forough Fazeli-Asl, Michael Minyi Zhang, and Lizhen Lin. A semi-Bayesian nonparametric estimator of the maximum mean discrepancy measure: Applications in goodness-of-fit testing and generative adversarial networks. *arXiv preprint arXiv:2303.02637*, 2023.
- [GAG21] Pierre Glaser, Michael Arbel, and Arthur Gretton. KALE flow: A relaxed KL gradient flow for probabilities with disjoint support. In M. Ranzato,

- A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8018–8031. Curran Associates, Inc., 2021.
- [GBG25] Alexandre Galashov, Valentin De Bortoli, and Arthur Gretton. Deep MMD gradient flow without adversarial training. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [GBR⁺12] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(1):723–773, 2012.
- [GM17] Jackson Gorham and Lester Mackey. Measuring sample quality with kernels. In D. Precup and Y. W. Teh, editors, *International Conference on Machine Learning*, pages 1292–1301. PMLR, PMLR, 2017.
- [HBM07] Zaïd Harchaoui, Francis Bach, and Eric Moulines. Testing for homogeneity with kernel Fisher discriminant analysis. In J. Platt, D. Koller, Y. Singer, and S. Roweis, editors, *Advances in Neural Information Processing Systems*, volume 20. Curran Associates, Inc., 2007.
- [HBSL24] Ye He, Krishna Balasubramanian, Bharath K. Sriperumbudur, and Jianfeng

- Lu. Regularized Stein variational gradient flow. *Foundations of Computational Mathematics*, pages 1–59, 2024.
- [HKSL24] Omar Hagrass, Bharath K. Sriperumbudur, and Bing Li. Spectral regularized kernel two-sample tests. *The Annals of Statistics*, 52(3):1076–1101, 2024.
- [HWAH24] Johannes Hertrich, Christian Wald, Fabian Altekrüger, and Paul Hagemann. Generative sliced MMD flows with riesz kernels. In *The Twelfth International Conference on Learning Representations*, 2024.
- [JKO98] Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.
- [KAFMA21] Anna Korba, Pierre-Cyril Aubin-Frankowski, Szymon Majewski, and Pierre Ablin. Kernel Stein discrepancy descent. In M. Meila and T. Zhang, editors, *International Conference on Machine Learning*, pages 5719–5730. PMLR, 2021.
- [KBGM25] Heishiro Kanagawa, Alessandro Barp, Arthur Gretton, and Lester Mackey. Controlling moments with kernel Stein discrepancies. *The Annals of Applied Probability*, 35(6):3818–3843, 2025.

- [KGBF25] Oscar Key, Arthur Gretton, François-Xavier Briol, and Tamara Fernandez. Composite goodness-of-fit tests with kernels. *Journal of Machine Learning Research*, 26(51):1–60, 2025.
- [KPLB26] Sherman Khoo, Dennis Prangle, Song Liu, and Mark Beaumont. Minimum distance summaries for robust neural posterior estimation. *arXiv preprint arXiv:2602.09161*, 2026.
- [MRW⁺99] Sebastian Mika, Gunnar Rätsch, Jason Weston, Bernhard Schölkopf, and Klaus-Robert Müller. Fisher discriminant analysis with kernels. In *Neural Networks for Signal Processing IX: Proceedings of the 1999 IEEE Signal Processing Society Workshop (Cat. No.98TH8468)*, pages 41–48, 1999.
- [MSR19] Youssef Mroueh, Tom Sercu, and Anant Raj. Sobolev descent. In Kamalika Chaudhuri and Masashi Sugiyama, editors, *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 2976–2985. PMLR, 2019.
- [OT11] Shin-ichi Ohta and Asuka Takatsu. Displacement convexity of generalized relative entropies. *Advances in Mathematics*, 228(3):1742–1787, 2011.

- [Ott01] Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. 2001.
- [SFL11] Bharath K. Sriperumbudur, Kenji Fukumizu, and Gert R. G. Lanckriet. Universality, characteristic kernels and RKHS embedding of measures. *Journal of Machine Learning Research*, 12(7):2389–2410, 2011.
- [SK25] Ziwei Su and Diego Klabjan. Differentiable calibration of inexact stochastic simulation models via kernel score minimization. *arXiv preprint arXiv:2411.05315*, 2025.
- [SKA⁺23] Antonin Schrab, Ilmun Kim, Mélisande Albert, Béatrice Laurent, Benjamin Guedj, and Arthur Gretton. MMD aggregated two-sample test. *arXiv preprint arXiv:2110.15073*, 2023.
- [SNRS26] Viktor Stein, Sebastian Neumayer, Nicolaj Rux, and Gabriele Steidl. Wasserstein gradient flows for moreau envelopes of f-divergences in reproducing kernel hilbert spaces. *Analysis and Applications*, 24(01):21–65, 2026.
- [SS19] Simo Särkkä and Arno Solin. *Applied stochastic differential equations*, volume 10. Cambridge University Press, 2019.

- [VW19] Santosh Vempala and Andre Wibisono. Rapid convergence of the unadjusted Langevin algorithm: Isoperimetry suffices. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.